

This video elucidates key concepts, mathematical derivations, and application problems regarding **Continuous Uniform Distributions**.

Summary

The Uniform Distribution, also known as the rectangular distribution, is characterized by a constant probability density across a specific interval.

Core principle:

Every point within the interval $([a, b])$ has equal likelihood of occurrence.

Key Takeaways

- **Probability Density Function (PDF):**
 $f(x) = \frac{1}{b-a}$
for $(a \leq x \leq b)$
- **Parameters:**
Two parameters define the distribution:
 - (a) (minimum value)
 - (b) (maximum value)
- **Mean:**
 $E[X] = \frac{a + b}{2}$
- **Variance:**
 $\text{Var}(X) = \frac{(b - a)^2}{12}$
- **Use Case:**
Applicable when outcomes are equally likely within known bounds.

1. Fundamental Principles of Uniform Distribution

A continuous uniform distribution exists when a random variable (X) is restricted to a finite interval $([a, b])$ and the probability is constant across that interval.

1.1 Derivation of the Probability Density Function

For a valid PDF:

$$\int_a^b f(x) dx = 1$$

Assume ($f(x) = k$) for ($a \leq x \leq b$).

$$\int_a^b k dx = 1$$

$$k[x]_a^b = 1$$

$$k(b - a) = 1$$

$$k = \frac{1}{b - a}$$

Therefore:

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$

1.2 Graphical Interpretation

The distribution forms a rectangle:

- Width = ($b - a$)
- Height = ($\frac{1}{b - a}$)

Total Area:

$$(b-a) \times \frac{1}{b-a} = 1$$

2. Statistical Properties and Parameters

The behavior of a uniform distribution depends entirely on (a) and (b).

Property	Formula	Description
Mean	$(\frac{a+b}{2})$	Midpoint of interval
Variance	$(\frac{(b-a)^2}{12})$	Spread around mean
Standard Deviation	$(\sqrt{\frac{(b-a)^2}{12}})$	Square root of variance
Parameters	(a, b)	Lower and upper bounds

3. Applied Problem Analysis

3.1 Parameter Recovery from Mean and Variance

Given:

Mean = 1

Variance = $\frac{4}{3}$

Using mean:

$$[\frac{a+b}{2} = 1]$$

$$[a + b = 2]$$

Using variance:

$$[\frac{(b-a)^2}{12} = \frac{4}{3}]$$

$$[(b-a)^2 = 16]$$

$$[b - a = 4]$$

Solving:

$$[a = -1, \quad b = 3]$$

Probability Calculation:

$$[P(X \leq 0) = \int_{-1}^0 \frac{1}{b-a} dx]$$

3.2 Probability in Quadratic Equations

Given:

$$[3x^2 + 6yx + (y + 6) = 0]$$

where $(y \sim U[1,6])$

Condition for real roots:

$$[D = b^2 - 4ac \geq 0]$$

Solve inequality in terms of (y) .

Probability:

$$[\frac{\text{Length of successful interval}}{b-a}]$$

3.3 Expected Waiting Time in Traffic System

Traffic cycle: 5 minutes total

- Green: $[0,2] \rightarrow$ waiting time = 0
- Red: $[2,5] \rightarrow$ waiting time = $(5 - x)$

Expected waiting time:

$$[E[\text{Wait}] = \int_0^2 0 \cdot f(x) dx + \int_2^5 (5-x) \cdot \frac{1}{5} dx]$$

Result: Expected waiting time = 0.9 minutes